

Experiment 1:

Comparison of Traditional Software Development and Low-Code Development

Aim:

To study and compare the Traditional Software Development approach with the Low-Code Development approach by understanding their development process, features, advantages, disadvantages, and applications.

Introduction to the Traditional Approach:

Definition:

The Traditional Software Development approach is the conventional method of building software where developers write source code manually using programming languages such as Java, Python, C#, PHP, or JavaScript. Every functionality is implemented through coding, testing, debugging, and deployment.

Working Process:

In this approach, the software development follows the Software Development Life Cycle (SDLC), where each phase is completed sequentially.

Features:

- Manual coding is required.
- Requires experienced software developers.
- High flexibility and customization.
- Suitable for large-scale enterprise applications.
- Development takes more time.

Advantages:

- Complete control over application design.
- Highly customizable.
- Better performance optimization.
- Suitable for complex applications.

Disadvantages:

- Time-consuming.
- Higher development cost.
- Requires skilled programmers.
- Longer maintenance cycle.

Applications:

- Banking systems
- Hospital Management Systems
- ERP Software
- E-commerce websites
- Enterprise Applications

Introduction to the Low-Code Approach**Definition:**

Low-Code Development is a modern software development approach where applications are built using graphical interfaces, drag-and-drop components, pre-built templates, and minimal manual coding.

Popular Low-Code Platforms include:

- Microsoft Power Apps
- Mendix
- Out Systems
- Appian
- Zoho Creator

Working Process:

Instead of writing code manually, developers use visual tools to design workflows, databases, forms, and business logic.

Features:

- Drag-and-drop interface
- Minimal coding
- Faster application development
- Reusable components
- Easy deployment

Advantages:

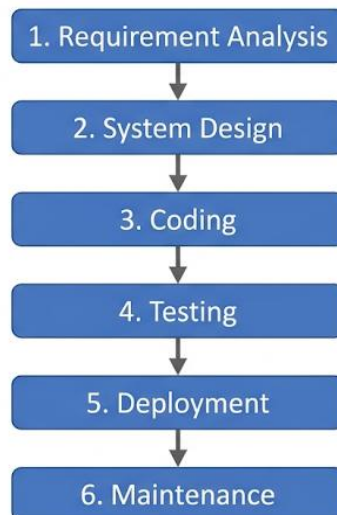
- Rapid application development
- Lower development cost
- Easy maintenance
- Suitable for non-programmers
- Faster deployment

Disadvantages:

- Limited customization
- Platform dependency
- Licensing costs
- Less suitable for highly complex applications

Applications:

- Business Process Automation
- Inventory Management
- Employee Management Systems
- CRM Applications
- Internal Business Applications

Diagram Showing Development Stages:**Traditional Software Development:****Traditional Software Development**

Low-Code Development:



Comparison Between Traditional and Low-Code Development:

Feature	Traditional Development	Low-Code Development
Development Method	Manual Coding	Visual Development
Programming Knowledge	Required	Minimal
Development Speed	Slow	Very Fast
Cost	High	Low
Customization	Very High	Moderate
Learning Curve	Difficult	Easy
Maintenance	Complex	Easy
Deployment	Manual	Automated
Best For	Large Complex Systems	Small and Medium Business Applications
Productivity	Lower	Higher

Result:

The Traditional Software Development approach provides complete flexibility and is suitable for complex enterprise applications but requires extensive coding, higher cost, and longer development time. The Low-Code Development approach significantly reduces development effort through visual tools, enabling faster application development with minimal coding. It is ideal for rapid business application development, though it offers less customization than traditional development.